Abstract

Globally, clinical mastitis (CM) is an important production disease in the dairy industry and has a great economic impact because of reduced milk yield, milk quality deterioration, treatment costs, involuntary culling, death, increased risk of antimicrobial resistance and reduced animal welfare. A cohort study was conducted on 24 randomly selected commercial dairy farms in Chattogram for six months (from May to October 2018) to (1) estimate bulk milk somatic cell count (BMSCC) (2) estimate the incidence rate of clinical mastitis (IRCM) at different levels (Cow, quarter and farm), (3) identify risk factors for BMSCC and for CM, (4) isolate pathogens causing CM, (5) determine minimum inhibitory concentration (MIC) of selected antibiotics against *Staphylococcus aureus* and (6) describe farmers perception and behavior and treatment strategies toward CM. Clinical mastitis was defined as grade-1 if there were changes only in milk, grade-2 for changes in milk and udder and grade-3 for changes in milk and udder along with systemic changes. Farm and animal level fixed and dynamic data were recorded. Milk samples were collected from the affected animals before administering any antimicrobial therapy. The BMSCC varied from 354×10^3 to $1,157\times10^3$ cells per mL of milk during three repeated counts on the selected farms. At the start of the study, in total 1,383 lactating cows were at risk and 204 cases of CM occurred during the study period. Estimated IRCM was 43.9 (95% CI: 38.2 to 50.5%) cases per 100 cows per year where the IRCM of grade-2 (25.5 cases per 100 cows per year) was higher than the IRCM of grade-1 (10.5 cases per 100 cows per year) and grade-3 (7.8 cases per 100 cows per year). At quarter level, the IRCM was 11.6 (95% CI: 10.2-13.3%) cases per 100 quarters per year where IRCM for grade-2 was also higher (6.5 cases per 100 quarters per year) than grade-1 (2.8 cases per 100 quarters per year) and grade-3 (2.4 cases per 100 quarters per year) similarly. The IRCM varied substantially between farms, ranging from 0 cases per 100 cows per year to 103.5 cases per 100 cows per year at cow level. This also varied at quarter level between farms ranging from 0 cases per 100 quarters per year to 52.7 cases per 100 quarters per year. In generalized linear regression model, BMSCC was positively correlated with animals purchased from other than own stock and had negatively correlated with dry floor condition. Using a generalized random effects logistic regression model the following factors were determined to be

associated with higher odds of CM: high milk yield (17 liters or more per day) versus lower milk yield (OR 2.8, 95% CI 1.9-4.1); early (OR 1.6, 95% CI 1.1-6.4) and mid (OR 1.2, 95% CI 0.8-2.0) lactation compared to late lactation, and finally cows with a body condition score (BCS) ≥3.26 (OR 1.5, 95% CI 1.0-2.2) compared to a BCS ≤3. Commonly isolated pathogens from cases of CM (n=167) were non-*aureus staphylococci* (24.0%), *Streptococcus* spp. (18.6%), *Bacillus* spp. (19.8%), *Staphylococcus aureus* (3.6%), *E. coli* (6.0%) and other Gram-negative bacteria (10.2%). Of all samples, 16.3% were culture negative. One methicillin resistant *Staphylococcus aureus* was detected and found resistant against penicillin, oxacillin, cefoxitin, gentamycin and tetracycline. Altogether, the IRCM in the Chattogram region of Bangladesh is high and *Staphylococcus* spp. are the most frequently isolated pathogens. Because of the large variation in IRCM between herds and the identification of several risk factors explaining this variation, it seems feasible to improve the existing situation by better farm management practices.

Keywords*:* Bulk milk somatic cell count, incidence rate of clinical mastitis, non-*aureus staphylococcus*, matrix assisted laser desorption/isonization time of flight, minimum inhibitory concentration